Distributed Quantum-Assisted Robust AoII Minimization in Satellite-Ground Integrated Edge Networks

Mohammad Arif Hossain, *Member, IEEE*, Tanzimul Alam Fahim, Weiqi Liu, *Member, IEEE*, and Nirwan Ansari, *Life Fellow, IEEE*

***Abstract*—Edge servers that act on a monitored process need an estimate that is correct, not merely recent. In satellite-ground integrated networks (SGINs), satellite handover and shadowing interrupt links for long enough that the monitored process can change state several times, so we adopt the age of incorrect information (AoII) as the measure of estimation quality. We formulate VANGUARD, a robust network-level problem that chooses update rates, serving-node association, and bandwidth to minimize the worst-case time-average AoII over a set of disruption scenarios, and solve it with SENTINEL, a distributed hybrid quantum-classical procedure. A renewal-interval decomposition turns the history-dependent AoII into a closed-form cost per inter-delivery interval; the minimax is dualized into classically updated scenario weights, so that each satellite-ground domain solves a linear assignment subproblem encoded as a compact quadratic unconstrained binary optimization (QUBO) problem whose one-hot structure is enforced by XY mixers; the domains are coordinated by ADMM, and every schedule is certified by its exact worst-case AoII. The evaluation uses Starlink ephemerides, the 3GPP TR 38.811 channel, acknowledgement feedback for all policies, an exact integer-programming reference, and paired tests over 30 to 60 seeds. The certificate held in 98.7% of orchestration epochs under real passes, the robust schedule's certificate is 2 to 7% below the nominal schedule's at no measurable cost in realized AoII, and SENTINEL with a state-aware trigger comes within 10 to 20% of the best per-slot heuristic while retaining the certificate. The compact formulation needs four to six times fewer qubits than a direct epigraph encoding, and thirteen domains of 8 to 38 qubits sampled on two IBM Heron processors decoded to the exact optimum; at these sizes the quantum sampler is correct but not a better optimizer than greedy association, and we state the scale at which that would change.**

***Index Terms*—6G, age of incorrect information (AoII), alternating direction method of multipliers (ADMM), mission-critical edge computing, non-terrestrial networks, quantum approximate optimization algorithm (QAOA), remote estimation, robust optimization, satellite-ground integrated network (SGIN)**

I. Introduction

Sixth-generation networks are expected to place decision-making at the edge [1]: an edge server that monitors a hazard indicator, the integrity of a structure, or the position of an asset must act on its own estimate of that process, often without a human in the loop. What matters to such a server is whether its estimate is still correct. Terrestrial mobile edge computing provides capacity and low delay where ground infrastructure exists, which is often not where monitoring is most needed, and satellite links extend coverage to the remaining regions at the price of intermittent connectivity [2]. The combination, a satellite-ground integrated network (SGIN), is the setting of this paper. We write SGIN rather than the more common SAGIN because the system has no aerial layer: there are no UAVs or high-altitude platforms, only satellites and ground nodes.

Satellite links fail in ways that terrestrial links rarely do. A low-earth-orbit satellite sets and hands the user over to the next one every few minutes; buildings and foliage block the line of sight for the length of a pass; shadow fading in the non-line-of-sight state exceeds ten decibels [3]. During any of these interruptions the monitored process keeps evolving, and by the time the link is restored the edge estimate may be wrong by several states rather than simply old. The age of information (AoI) [4], [5] measures only elapsed time and cannot tell these two situations apart. The age of incorrect information (AoII) [6], [7] can: it is zero while the estimate is correct and grows with the duration and the magnitude of a mismatch. AoII has been analyzed mainly for one source and one monitor on a point-to-point channel. Carrying it to a network raises two difficulties: the metric depends on the whole trajectory of the source since the estimate was last correct, so it does not fit the per-slot cost structure of the assignment problems used for SGIN resource allocation; and the duration of a mismatch is governed by outages of uncertain length, so an allocation must be judged by its worst case over plausible disruptions rather than by its average.

We treat the problem as robust combinatorial optimization. A renewal-interval decomposition shows that a Markov source with known transition matrix accumulates, between two deliveries, an expected AoII that depends only on the length of the interval and can be tabulated offline, so the channel enters the objective through a single number per scenario and the resulting robust scheduling problem, VANGUARD, is a minimax assignment problem over binary variables. The question is then how such a problem can be given to a quantum sampler. A direct encoding of the minimax, with an epigraph register and a slack register per scenario on top of every admissible decision variable, needs four to six times the

Mohammad Arif Hossain is with Department of Engineering Technology and Quantum Research Interdisciplinary Science and Education (QRISE) Center at Middle Tennessee State University (MTSU), TN 37132 USA (e-mail: Mohammad.Hossain@mtsu.edu), Tanzimul Alam Fahim is with the Computational Science PhD Program at MTSU, Murfreesboro, TN 37132 USA, Weiqi Liu is with the Department of Computer Science and Computer Information Systems, Auburn University at Montgomery, AL 36117 USA, and Nirwan Ansari is with the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, NJ 07102 USA.

qubits of the encoding proposed here and produces a penalty landscape on which classical and quantum local search alike fail to find a feasible point; we document this in Section VI. SENTINEL, for Satellite-ground Edge Network orchestration for Trustworthy Information via Node-level Estimation Limits, instead dualizes the minimax into scenario weights that a classical outer loop updates, so that each inner subproblem is linear and is encoded as a compact QUBO over the decision variables and a few capacity-slack bits; the one-hot constraint of each user is enforced by the XY mixer of the circuit rather than by a penalty [8]. Satellite-ground domains are coordinated by the sharing form of ADMM [9], and every assembled schedule is evaluated exactly under every scenario, so the reported worst case is a certificate that does not rest on the solver's optimality. The procedure is hybrid at every level: a classical optimizer tunes the circuit angles; the scenario weights, the coordination, the repair, and the certificate are classical; and domains beyond the sampler's capacity are solved by a classical annealer on the same QUBO. Section VI states which solver produced each result.

The evaluation is built so that a reader can trust it: Starlink ephemerides propagated with SGP4 [10], the suburban channel of 3GPP TR 38.811 [3], acknowledgement feedback for every policy, an exact integer-programming reference, common random episodes with paired tests, and runs of the quantum step on two IBM Heron processors. Its findings are modest and precisely stated. The worst-case certificate held in 98.7% of orchestration epochs under real passes; the robust schedule's certificate is 2 to 7% below its nominal counterpart's in 67 to 100% of seeds, while the two are statistically indistinguishable in realized AoII, so the value of the scenario set is the guarantee, not a lower average. A per-slot policy that transmits whenever its observed AoII is positive is the strongest heuristic when feedback is available and coincides with the Whittle-index scheduler [11], [12] at these bandwidths; SENTINEL with the same trigger comes within 10 to 20% of it while keeping the certificate, and is 25 to 30% better than it at 15 to 20 users without feedback. The compact formulation matches the exact optimum on the 10- and 25-UE instances with four to six times fewer qubits than the epigraph encoding, and thirteen domains of 8 to 38 qubits sampled on `ibm_kingston` and `ibm_fez` decoded to the exact optimum. On every instance that fits a current device greedy association is already optimal, so the quantum step is a correct, hardware-runnable constrained sampler and not yet a better optimizer; we quantify this per shot against a uniform-sampling baseline and state the scale at which it would change.

The contributions of this paper are the following.

- The robust scheduling problem VANGUARD, which minimizes the worst-case time-average AoII of a network of monitored edge nodes over a set of satellite shadowing and handover scenarios, jointly over update rates, association, and bandwidth, under capacity, visibility, rate, and AoII-budget constraints.
- A renewal-interval decomposition that expresses the expected AoII accumulated between deliveries in closed form and reconciles the history dependence of AoII with binary-variable optimization; verified against Monte Carlo simulation to within one percent.
- SENTINEL, a distributed hybrid quantum-classical procedure that dualizes the minimax into classically updated scenario weights, encodes each domain's subproblem as a compact QUBO with XY-mixer-enforced one-hot structure, prunes dominated candidates, and coordinates the domains by ADMM with a warm start and a cost-scaled penalty, with a certificate of the worst-case AoII of every schedule.
- An evaluation under Starlink ephemerides and the TR 38.811 channel against an exact reference, with paired statistics, Whittle-index and trained learning baselines, an ablation of failure feedback, sensitivity studies, and runs of the quantum step on two superconducting processors with its scale limit stated explicitly.

Section II reviews related work. Section III presents the system model, Section IV formulates VANGUARD and its binary reformulations, Section V develops the dualized compact QUBO, the XY-mixer QAOA, and the ADMM coordination, Section VI reports the evaluation, and Section VII concludes.

II. State of the Art

1) Satellite-Ground Integrated Edge Computing: Mobile edge computing [13] has been extended beyond terrestrial coverage by placing edge servers on satellites and base stations [14]–[16], with task offloading under deterministic or average channel assumptions as the dominant problem [17], [18] and completion delay as the objective; the correctness of what the server believes about the monitored process has not been the quantity optimized. The 3GPP study on non-terrestrial networks [3] provides elevation-dependent line-of-sight probabilities, clutter losses, and shadow-fading statistics, and SGP4 with public two-line elements [10] gives pass geometry for any constellation; most SGIN evaluations use neither, although with real passes the dominant disruptions are handovers and non-line-of-sight passes rather than coverage gaps.

2) Age of Information and Age of Incorrect Information: AoI is a widely studied but content-agnostic timeliness measure [4], [5], [12], [19], [20]. AoII [6] penalizes the duration and magnitude of estimation error and has been analyzed for a single source under threshold and semi-Markov policies [7], [21] and over unreliable, power-constrained channels [22]. For several sources over shared channels the standard tool is the Whittle index of the restless bandit [11], applied to AoI by Kadota et al. [12], and we include it as a baseline. Missing is a formulation in which association, bandwidth, and update rates are chosen jointly across heterogeneous serving nodes, under outages that span several source transitions, with a guarantee on the worst case.

3) Robust Allocation and Learning-Based Scheduling: Robust optimization minimizes cost against the worst realization of an uncertain parameter in a bounded set [23], [24]. When the uncertainty set is a finite list of scenarios, the minimax is a maximization over scenario weights in the simplex, and the saddle point can be approached by multiplicative-weights updates [25]; this is the mechanism we use to keep the quantum

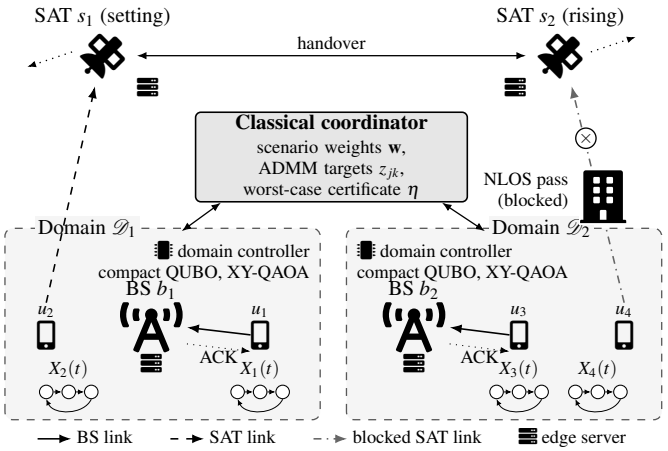


Fig. 1. System model: two satellite-ground domains with their base stations, shared LEO satellites, monitored sources $X_i(t)$, and the classical coordinator. Solid: BS link with acknowledgement (dotted); dashed: satellite link; dash-dotted: satellite link blocked during a non-line-of-sight pass.

subproblem linear. Robust allocation in satellite-ground edge computing has treated uncertainty as a perturbation of rate or delay, not as the driver of estimation error. Deep reinforcement learning [26], [27] has been applied to AoI- and AoII-aware scheduling in small systems; it optimizes an expected return and gives no bound on the estimation error under disruptions outside its training distribution. That does not disqualify it as a baseline, and we train one with the same reward, feedback, and episodes as every other policy.

4) Quantum-Assisted Combinatorial Optimization: QUBO formulations [28] solved by QAOA [29] or by annealing [30] have been proposed for channel assignment and scheduling in wireless networks [31]. Two developments bear on constrained assignment: the alternating-operator form of QAOA [8], whose mixer preserves a constraint subspace so that one-hot constraints need no penalty terms, and the concentration of optimal QAOA parameters across instances [32], which allows angles optimized on a small instance to be reused on a larger one; error suppression by dynamical decoupling [33] and randomized compiling [34] is inexpensive, and we report its effect. QAOA has not been applied to a correctness objective, and reports of its use for network resource allocation rarely state at what size the quantum sampler stops being competitive with a classical heuristic.

5) Research Gap: No existing work formulates AoII as a robust minimax multi-node scheduling problem with joint association and resource allocation, targets information correctness rather than delay in robust SGIN allocation, or applies quantum-assisted optimization to a correctness objective with an encoding a current device can execute. SENTINEL addresses these three points.

III. System Model

We consider a satellite-ground integrated edge network in which monitoring edge nodes (UEs) observe time-varying physical processes and transmit status updates to edge servers

TABLE 1. Key Notations

Symbol	Definition
$\mathcal{U}$, $\mathcal{S}$, $\mathcal{B}$	Sets of UEs, SATs, and BSs
α_{ij}, ω_{in}	Association of u_i with s_j, b_n
β_{ij}^{us}, β_{in}^{ub}	Bandwidth allocated to u_i by s_j, b_n
B_j^s, B_n^b	Bandwidth capacity of s_j, b_n
$R_{ij}^{us}(\ell)$, R_{in}^{ub}	Uplink rate of u_i to s_j (scenario ℓ), to b_n
$V_{ij}(t)$, Θ	Elevation of s_j at u_i; minimum elevation
$\Delta_i^{II}(t)$, $\Delta_i^{II,\max}$	Instantaneous AoII; AoII budget
$\Psi_i(Y)$	Expected AoII accumulated over an interval of length Y
ρ_i, $\bar{\rho}_i^h$, $\bar{\beta}_m^q$	Update rate of u_i; h-th rate level (H_i levels); q-th bandwidth level of node m (Q_m levels)
x_{imqh}	1 if u_i uses node m, bandwidth level q, rate level h
$\mathscr{A}_i$, $\mathscr{C}_i$, $\varkappa$	Admissible combinations of u_i; pruned candidates; candidates kept per UE
$\psi_{imqh}^{(\ell)}$	Time-average AoII cost under scenario ℓ
$\mathbf{w}$, μ, η	Scenario weights; weight-update step; worst-case AoII
z_{jk}, υ_{jk}, λ_c	ADMM bandwidth target, scaled dual, penalty

hosted at terrestrial base stations (BSs) or at visible low-earth-orbit satellites (SATs). Let $\mathcal{U}$, $\mathcal{S}$, and $\mathcal{B}$ denote the sets of UEs, SATs, and BSs, indexed by u_i, s_j, and b_n, and let $\mathscr{M} = \mathcal{S} \cup \mathcal{B}$ be the set of serving nodes. Each serving node hosts an edge server that keeps a running estimate of every process assigned to it and acts on that estimate. Time is slotted with slot duration T_s. UEs have low mobility over a decision interval; the satellite topology changes continuously because of orbital motion. Table 1 lists the notation.

In contrast to the task-offloading models that dominate SGIN edge computing, the quantity of interest here is the correctness of the server's estimate. It depends on the dynamics of the monitored process, on when the UE samples and transmits, and on whether the transmission is delivered before the process moves again. Satellite handover and shadowing make the last of these uncertain, and an allocation chosen under nominal channel conditions can leave the server with an estimate that is wrong rather than late.

The network is partitioned into K local satellite-ground domains $\mathcal{D}_k = (\mathcal{U}_k, \mathcal{S}_k, \mathcal{B}_k)$, $k \in \mathcal{K} = \{1, \ldots, K\}$, where the UE subsets partition $\mathcal{U}$ and a SAT may be visible to several domains. Each domain has a classical controller that builds and solves a local robust binary problem with a hybrid QAOA solver; a classical coordinator reconciles the bandwidth of SATs shared across domains and certifies every assembled network schedule by its exact worst-case AoII.

A. Satellite Geometry and Communications Model

Satellite positions are obtained from a constellation ephemeris; the evaluation propagates a public two-line element set with SGP4 [10], but the model only requires, for every SAT s_j and UE u_i, the slant range $d_{ij}^{us}(t)$ and the elevation $V_{ij}(t)$ in each slot. A SAT can serve u_i in slot t only if $V_{ij}(t) \geq \Theta$. A UE keeps its serving SAT until it sets below Θ and is then handed over to the SAT with the highest elevation, so a dense constellation yields passes of several minutes separated by handover interruptions and no coverage gaps at the masks considered here.

The nominal path loss (in dB) between u_i and s_j is

$$\varphi_{ij}^{us} = \varsigma^{us} + \zeta^{us}\log_{10}\big(f_c^{us}\big) + \xi^{us}\log_{10}\Big(d_{ij}^{us}\Big) + \lambda^{\mathrm{atm}}(V_{ij}), \quad (1)$$

with carrier frequency f_c^{us}, the reference loss and exponents ς^{us}, ζ^{us}, ξ^{us} of free-space propagation, and the atmospheric absorption $\lambda^{\mathrm{atm}}(V) \propto 1/\sin V$. The UE-BS path loss φ_{in}^{ub} uses the ground distance d_{in}^{ub}, a terrestrial exponent $\xi^{ub} > 2$, and a fixed margin. With transmit power p_i and antenna gains G^u and G^s at the terminal and the satellite, the nominal uplink rate with bandwidth β_{ij}^{us} is

$$R_{ij}^{us} = \beta_{ij}^{us}\,\min\left\{\log_2\left(1 + \frac{10^{(p_i+G^u+G^s-\varphi_{ij}^{us})/10}}{N_0\,\beta_{ij}^{us}\,F^s}\right),\ \bar{\gamma}\right\}, \quad (2)$$

where N_0 is the noise power spectral density, F^s the receiver noise figure, and $\bar{\gamma}$ the spectral efficiency of the highest modulation and coding scheme; R_{in}^{ub} is defined in the same way with G^b. Each node divides its bandwidth among the UEs it serves, so a link's rate does not depend on other UEs' association once the bandwidth level is fixed. Co-channel interference is absorbed in the noise figure, and residual Doppler, which the NTN frequency pre-compensation removes to within the subcarrier spacing, is not modelled.

Large-scale channel. Each satellite link also carries a large-scale state drawn once per pass from the TR 38.811 model [3]: with the elevation-dependent probability $p_{\mathrm{LOS}}(V)$ the pass is in line of sight, otherwise it carries the clutter loss $\lambda^{\mathrm{cl}}(V)$ of the band (17 to 25 dB at Ku band and suburban elevations); shadow fading with the state-dependent deviation $\sigma_{\mathrm{sf}}(V)$ is drawn per pass and small-scale fading per slot. About one pass segment in twenty is in non-line-of-sight in the suburban environment of Section VI, and such a pass blocks the link for its whole duration at any bandwidth level.

B. Monitored Source and Estimation Model

Each UE u_i observes a process $X_i(t)$ modeled as a discrete-time Markov chain on a finite state space $\mathcal{X}_i$ with transition matrix P_i, in which the states are ordered severity or status levels. The UE controls a generation indicator $g_i(t) \in \{0, 1\}$ that specifies whether an update is sampled and transmitted in slot t. We assume that T_s exceeds the transmission and propagation delay of a successful update, so that an update sampled at the start of slot t is delivered within that slot or lost. Let $t_{i,\kappa}$ be the κ-th slot in which an update from u_i is delivered and $Y_{i,\kappa} = t_{i,\kappa+1} - t_{i,\kappa}$ the inter-delivery interval. The server assigned to u_i holds the estimate

$$\hat{X}_i(t) = X_i(t_{i,\kappa}), \qquad t_{i,\kappa} \le t < t_{i,\kappa+1}, \quad (3)$$

which is fixed between deliveries while the process keeps moving. With $\Lambda_i(t) = \max\{\tau \le t : X_i(\tau) = \hat{X}_i(\tau)\}$ the most recent slot in which the estimate was correct, the instantaneous AoII is [6]

$$\Delta_i^{II}(t) = \big(t - \Lambda_i(t)\big)\, f_i\big(X_i(t), \hat{X}_i(t)\big), \quad (4)$$

where $f_i(x, \hat{x}) \ge 0$ with $f_i(x, x) = 0$ measures the severity of a mismatch; we use $f_i(x, \hat{x}) = |x - \hat{x}|$. AoII returns to zero without a delivery whenever the process revisits the estimated state.

C. Disruption Scenarios

The orchestrator plans against a finite scenario set $\mathcal{L} = \{1, \dots, L\}$. Scenario ℓ adds a loss $\delta_{ij}^{us}(\ell)$ to every UE-SAT link for a window of $T^{\mathrm{dis}}(\ell)$ slots,

$$\tilde{\varphi}_{ij}^{us}(\ell) = \varphi_{ij}^{us} + \delta_{ij}^{us}(\ell), \quad (5)$$

the scenario rate $R_{ij}^{us}(\ell)$ follows from (2), and an update sampled in slot t is delivered under scenario ℓ only if $R_{ij}^{us}(\ell)\,T_s \ge D_i$ and $V_{ij}(t) \ge \Theta$, where D_i is the update size in bits. BS links are unaffected. We use five scenarios: the nominal channel; a deep shadowing event in which every satellite link loses σ_{sf} of the non-line-of-sight state, 12 dB in the suburban tables, for a window of T^{dis} slots; the handover interruption of each individual satellite, a complete outage of that link for the duration of a handover; and the joint outage of all satellites for the deep-shadowing window. A mild 6 dB scenario was dropped after the sensitivity study of Section VI showed that it never binds; the handover scenarios are kept per satellite because the hedge appears only once the second shared satellite's handover is in the set.

A scenario thus determines the length of the outage window and therefore the inter-delivery interval $Y_{i,\kappa}(\ell)$, the number of slots from $t_{i,\kappa}$ until the next slot in which $g_i(t) = 1$ and the delivery condition holds; a longer window lets $X_i(t)$ pass through more states. The scenario set is the planner's model of the disruption process, whereas the simulator of Section VI draws its disruptions from the TR 38.811 state process and the ephemeris; the agreement between the two is what the certificate's validity measures.

D. Renewal-Interval AoII Decomposition

Because $\Delta_i^{II}(t)$ depends on the trajectory of $X_i(t)$ since the estimate was last correct, it is not a per-slot function of the channel state and cannot be used directly as the cost coefficient of a binary variable. The delivery process is, however, a renewal process. Under the same-slot delivery assumption the estimate is exact at each delivery, so $\Delta_i^{II}(t_{i,\kappa}) = 0$, and between two deliveries the estimate is fixed at $\hat{x} = X_i(t_{i,\kappa})$ while the AoII evolves under the source's own dynamics, independently of the channel.

Let $\tilde{P}_{i,\hat{x}} = P_i\big(\mathbf{I} - \mathbf{e}_{\hat{x}}\mathbf{e}_{\hat{x}}^{\top}\big)$ be the taboo matrix with the $\hat{x}$-th column of P_i set to zero, so that $[\tilde{P}_{i,\hat{x}}^{a}]_{\hat{x}y}$ is the probability of moving from $\hat{x}$ to y in ι steps without returning to $\hat{x}$, and let $\mathbf{f}_{i,\hat{x}}$ be the vector with entries $f_i(y, \hat{x})$. The AoII s slots after a delivery equals $\iota\, f_i(y, \hat{x})$ exactly when the source was last at $\hat{x}$ at slot $s - \iota$ and has since moved to y without returning, so

$$\mathbb{E}\big[\Delta_i^{II}(t_{i,\kappa} + s)\,\big|\,\hat{x}\big] = \sum_{\iota=1}^{s} \iota\,\big[P_i^{s-\iota}\big]_{\hat{x}\hat{x}}\big[\tilde{P}_{i,\hat{x}}^{\iota}\,\mathbf{f}_{i,\hat{x}}\big]_{\hat{x}}, \quad (6)$$

and the expected AoII accumulated over an interval of length Y is

$$\Psi_i(Y \mid \hat{x}) = \sum_{s=1}^{Y-1} \mathbb{E}\big[\Delta_i^{II}(t_{i,\kappa} + s)\,\big|\,\hat{x}\big],$$
$$\Psi_i(Y) = \sum_{\hat{x}\in\mathcal{X}_i} \pi_i(\hat{x})\,\Psi_i(Y \mid \hat{x}), \quad (7)$$

where π_i is the stationary distribution of P_i. Averaging over π_i is legitimate because the update decisions do not depend on the realized source state, so the delivered state is stationary. By the renewal-reward theorem the time-average AoII of u_i when its deliveries are Y slots apart is $\Psi_i(Y)/Y$. Both quantities depend only on Y and P_i and are tabulated once per UE for $Y \le Y^{\max}$; the table agrees with Monte Carlo simulation of the source and delivery process over 1.5×10^6 slots within one percent at every interval length used.

Equation (7) separates the source from the channel. The only channel-dependent quantity the planner must reason about is the inter-delivery interval, which under scenario ℓ is deterministic for a given allocation and is denoted $Y_i(\ell)$; the cost is the tabulated ratio $\Psi_i(Y_i(\ell))/Y_i(\ell)$.

E. Binary Decision Model

The update rates and bandwidths are continuous. We discretize the bandwidth of each node $m \in \mathscr{M}$ into Q_m levels $\{\bar{\beta}_m^1, \dots, \bar{\beta}_m^{Q_m}\}$ and the update rate of u_i into H_i levels $\{\bar{\rho}_i^1, \dots, \bar{\rho}_i^{H_i}\} \subset [\rho_i^{\min}, \rho_i^{\max}]$, and define

$$x_{imqh} = \begin{cases} 1, & \text{if } u_i \text{ uses node } m, \text{ level } q, \text{ rate } h, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

The original decisions are recovered as $\alpha_{ij} = \sum_{q,h} x_{ijqh}$, $\omega_{in} = \sum_{q,h} x_{inqh}$, $\beta_{ij}^{us} = \sum_{q,h} \bar{\beta}_j^q x_{ijqh}$, $\beta_{in}^{ub} = \sum_{q,h} \bar{\beta}_n^q x_{inqh}$, and $\rho_i = \sum_{m,q,h} \bar{\rho}_i^h x_{imqh}$. The scenario does not index the variable, because it is realized by the environment rather than chosen; each variable instead carries L precomputed cost coefficients. After the inadmissible combinations are removed (Section IV-A) the number of decision variables is

$$N_x = \sum_{i \in \mathscr{U}} |\mathscr{A}_i| \;\le\; \sum_{i \in \mathscr{U}} \sum_{m \in \mathscr{M}} Q_m H_i, \tag{9}$$

which grows linearly with the number of UEs, nodes, and levels. Section V prunes each $\mathscr{A}_i$ further to a candidate set $\mathscr{C}_i$ of the $\varkappa$ combinations with the smallest worst-case cost, which in our instances never changed the optimum and reduced the qubit count by a further factor of two to three.

F. Delivery Feedback and Link-Failure Handling

A serving node acknowledges each delivered update. A UE whose transmissions to a node have failed in every slot of a window of T^{f} slots declares the link failed and avoids that node for T^{b} slots or, for a satellite, until its next handover; its controller re-solves the domain problem without that link at the next orchestration interval. The rule applies to every policy in Section VI, since any state-aware baseline needs the same information to be competitive; both regimes are reported.

G. Distributed Orchestration Model

Each domain $\mathscr{D}_k$ holds the local decision vector $\mathbf{x}_k = \{x_{imqh} \mid u_i \in \mathscr{U}_k, (m,q,h) \in \mathscr{C}_i\}$ and the local state $\mathbf{s}_k = [\mathbf{\Psi}_k, \mathbf{V}_k, \mathbf{X}_k, \mathbf{B}_k, \mathbf{E}_k, \mathbf{\Xi}_k]$, with $\mathbf{\Psi}_k$ the tabulated AoII costs of its UEs, $\mathbf{V}_k$ the elevations, $\mathbf{X}_k$ the transition matrices and budgets, $\mathbf{B}_k$ the bandwidth levels, $\mathbf{E}_k$ the ephemeris over the decision interval, and $\mathbf{\Xi}_k$ the scenario set restricted to the domain. The controller re-solves whenever a satellite becomes visible or sets, when a link failure is reported, and at least once per re-solve period. Within a solve, a classical outer loop maintains scenario weights, each domain solves a linear assignment subproblem on a quantum sampler (or, when the domain exceeds the sampler's capacity, on a classical annealer with the same QUBO), and the coordinator runs the sharing form of ADMM on the shared satellites' bandwidth and evaluates the worst-case AoII of each assembled schedule exactly. The orchestration latency extends every inter-delivery interval by ϵ_i slots and enters the cost through $\Psi_i(Y_i(\ell) + \epsilon_i)$; in the evaluation it is measured, not assumed.

IV. Problem Formulation

The objective is to minimize the worst-case time-average AoII across all monitored UEs over the scenario set, by jointly choosing update rates, association, and bandwidth. Let $\boldsymbol{\chi}_i = (\rho_i, \{\alpha_{ij}\}_j, \{\omega_{in}\}_n, \{\beta_{ij}^{us}\}_j, \{\beta_{in}^{ub}\}_n)$ collect the decisions of u_i and $Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})$ its inter-delivery interval under realization $\boldsymbol{\xi} \in \Omega$. The robust Volatility-Aware Network-wide Generation, Update, Association, and Resource orchestration under Disruption (VANGUARD) problem is

$$\begin{aligned}
\mathscr{P}_1 : \; & \min_{\{\boldsymbol{\chi}_i\}_{i \in \mathscr{U}}} \; \max_{\boldsymbol{\xi} \in \Omega} \; \frac{1}{|\mathscr{U}|} \sum_{i \in \mathscr{U}} \frac{\Psi_i\big(Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})\big)}{Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})} \\
\text{s.t. } & \mathrm{C1} : \sum_{j \in \mathscr{S}} \alpha_{ij} + \sum_{n \in \mathscr{B}} \omega_{in} = 1, \; \forall\, i \in \mathscr{U}, \\
& \mathrm{C2} : \sum_{i \in \mathscr{U}} \alpha_{ij}\, \beta_{ij}^{us} \le B_j^s, \; \forall\, j \in \mathscr{S}, \\
& \mathrm{C3} : \sum_{i \in \mathscr{U}} \omega_{in}\, \beta_{in}^{ub} \le B_n^b, \; \forall\, n \in \mathscr{B}, \\
& \mathrm{C4} : \frac{\Psi_i\big(Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})\big)}{Y_i(\boldsymbol{\chi}_i, \boldsymbol{\xi})} \le \Delta_i^{II,\max}, \; \forall\, i \in \mathscr{U}, \; \forall\, \boldsymbol{\xi} \in \Omega, \\
& \mathrm{C5} : \alpha_{ij}\big(V_{ij} - \Theta\big) \ge 0, \; \forall\, i \in \mathscr{U}, \; \forall\, j \in \mathscr{S}, \\
& \mathrm{C6} : \rho_i^{\min} \le \rho_i \le \rho_i^{\max}, \; \forall\, i \in \mathscr{U}, \\
& \mathrm{C7} : \alpha_{ij}, \; \omega_{in} \in \{0,1\}, \; \forall\, i, j, n.
\end{aligned}$$

C1 routes each UE through exactly one serving node per decision interval. C2 and C3 are the bandwidth capacities. C4 is the correctness constraint: the expected time-average AoII of every UE stays within its budget $\Delta_i^{II,\max}$ under every realization, including the worst. It is stated in expectation because the instantaneous AoII of a Markov source cannot be bounded deterministically; the source may stay in a mismatched state for a whole interval. C5 restricts association to visible satellites, C6 bounds the update rate by the freshness requirement and the UE's energy or channel-access budget, and C7 makes association binary.

$\mathscr{P}_1$ is a mixed-integer nonlinear minimax program. With a single scenario and fixed rates it contains the generalized assignment problem and is NP-hard. The inner maximization adds a difficulty of its own: because Ψ_i is a nonlinear, source-specific function of the interval, the realization that maximizes the network AoII is not necessarily the one with the worst channel but depends on how each source's volatility interacts with the outage it induces; and the association set defined by C5 changes at every decision interval as satellites move.

A. Robust Binary Reformulation

With the discretization of Section III-E and the variable x_{imqh} of (8), each scenario ℓ gives, for every UE-node pair and bandwidth level, a rate $R_{imq}(\ell)$ from (2) with the scenario path loss. The link is in outage throughout the scenario window if that rate cannot deliver an update within a slot, so the induced outage is

$$\tau_{imq}(\ell) = \begin{cases} T^{\mathrm{dis}}(\ell)\,\mathbb{I}\{R_{imq}(\ell)\,T_s < D_i\}, & m \in \mathcal{S}, \\ 0, & m \in \mathcal{B}. \end{cases} \tag{10}$$

The bandwidth level matters: a larger allocation can keep the rate above the delivery threshold through deep shadowing, whereas a handover interruption is an outage at any bandwidth. A UE whose update fails during an outage transmits a fresh sample as soon as the link returns. The worst case is an outage that begins at a scheduled sampling slot, which gives

$$Y_{imqh}(\ell) = \left\lceil \frac{1}{\bar{\rho}_i^h T_s} \right\rceil + \tau_{imq}(\ell), \qquad \psi^{(\ell)}_{imqh} = \frac{\Psi_i\big(Y_{imqh}(\ell)\big)}{Y_{imqh}(\ell)}, \tag{11}$$

precomputed for every (i, m, q, h, ℓ); with orchestration overhead, $Y_{imqh}(\ell) + \epsilon_i$ replaces $Y_{imqh}(\ell)$.

Pre-elimination. The admissible set $\mathcal{A}_i$ consists of the combinations (m, q, h) for which (i) $m \in \mathcal{B}$, or $m \in \mathcal{S}$ with $V_{im} \geq \Theta$; (ii) the nominal rate satisfies $R_{imq}T_s \geq D_i$; and (iii) $\max_\ell \psi^{(\ell)}_{imqh} \leq \Delta_i^{II,\max}$. Condition (i) enforces C5 and (iii) is the discrete form of C4: since each UE selects exactly one combination, it meets its budget under every scenario if and only if its selection lies in $\mathcal{A}_i$, so the budget is enforced by removing variables rather than by penalty terms, at no cost in qubits. A UE with empty $\mathcal{A}_i$ (a satellite-only UE whose budget lies below the AoII of a full handover outage) is assigned its least-bad combination and reported as a budget violation.

The binary reformulation of VANGUARD is

$$\begin{aligned} \mathscr{P}_2 :\ & \min_{\mathbf{x}} \ \max_{\ell \in \mathscr{L}} \ \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \sum_{(m,q,h) \in \mathcal{A}_i} \psi^{(\ell)}_{imqh}\, x_{imqh} \\ \text{s.t. } & \mathrm{D1}: \sum_{(m,q,h) \in \mathcal{A}_i} x_{imqh} = 1, \ \forall\, i \in \mathcal{U}, \\ & \mathrm{D2}: \sum_{i \in \mathcal{U}} \sum_{(j,q,h) \in \mathcal{A}_i} \tilde{\beta}^q_j\, x_{ijqh} \leq B^s_j, \ \forall\, j \in \mathcal{S}, \\ & \mathrm{D3}: \sum_{i \in \mathcal{U}} \sum_{(n,q,h) \in \mathcal{A}_i} \tilde{\beta}^q_n\, x_{inqh} \leq B^b_n, \ \forall\, n \in \mathcal{B}, \\ & \mathrm{D4}: x_{imqh} \in \{0,1\}, \ \forall\, i \in \mathcal{U}, \ (m,q,h) \in \mathcal{A}_i. \end{aligned}$$

D1 is the one-hot selection, D2 and D3 the discrete capacities, D4 the binary domain.

B. Epigraph Form, Certificate, and Exact Reference

Introducing the worst-case value η gives the equivalent epigraph form

$$\begin{aligned} \mathscr{P}_3 :\ & \min_{\mathbf{x},\,\eta} \ \eta \\ \text{s.t. } & \mathrm{E1}: \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \sum_{(m,q,h) \in \mathcal{A}_i} \psi^{(\ell)}_{imqh}\, x_{imqh} \leq \eta, \ \forall\, \ell \in \mathscr{L}, \\ & \mathrm{D1\text{–}D4}, \end{aligned}$$

in which E1 is tight for at least one scenario at the optimum. $\mathscr{P}_3$ serves two purposes in this paper, neither of which is to be solved on a quantum device. First, it is the certificate: for any schedule $\mathbf{x}$ returned by any solver, the L left-hand sides of E1 are evaluated exactly, so the worst-case time-average AoII of that schedule over $\mathscr{L}$ is known with certainty. Second, $\mathscr{P}_3$ is a mixed-integer linear program, and up to 75 UEs it is solved to proven optimality by a branch-and-cut solver [35] in well under a second, with a time limit reached only on some 100-UE instances; this gives an exact reference for every heuristic and quantum solver we compare. A direct QUBO encoding of $\mathscr{P}_3$, with η on a binary grid and a slack register for each of the L constraints E1, was the encoding of an earlier version of this work; it multiplies the qubit count by four to six and produces a penalty landscape in which neither simulated annealing nor QAOA found a feasible assignment on instances of ten UEs. Section V therefore works with $\mathscr{P}_2$ directly, replacing the inner maximization by scenario weights so that every quantum subproblem is linear in $\mathbf{x}$ and the only auxiliary qubits are the slack bits of the capacity constraints D2 and D3.

V. Proposed Solution

Problem $\mathscr{P}_2$ is a robust assignment problem whose only coupling across UEs is the capacity of the serving nodes. SENTINEL solves it in three steps. The minimax over scenarios is handled by a classical outer loop on scenario weights, so that every subproblem handed to a quantum sampler has a linear objective. Each domain's subproblem is encoded as a compact QUBO over its decision variables and a few capacity-slack bits, with the one-hot constraint enforced by the mixer of the quantum circuit rather than by a penalty. The domains are coordinated through the sharing form of ADMM. Every schedule produced at any point of the procedure is certified exactly by its worst-case AoII over $\mathscr{L}$, so the procedure can be stopped at any time with a valid guarantee in hand. The procedure is hybrid at every level, as Section I describes, and Section VI states which inner solver produced each result.

A. Scenario-Weight Dualization of the Minimax

Write $c^{(\ell)}(\mathbf{x}) = \frac{1}{|\mathcal{U}|} \sum_i \sum_{(m,q,h) \in \mathcal{A}_i} \psi^{(\ell)}_{imqh} x_{imqh}$ for the network AoII under scenario ℓ. For any $\mathbf{x}$ and any weight vector $\mathbf{w}$ in the simplex $\Delta_L = \{\mathbf{w} \geq 0,\ \sum_\ell w_\ell = 1\}$,

$$\max_{\ell \in \mathscr{L}} c^{(\ell)}(\mathbf{x}) = \max_{\mathbf{w} \in \Delta_L} \sum_{\ell \in \mathscr{L}} w_\ell\, c^{(\ell)}(\mathbf{x}), \tag{12}$$

so $\mathscr{P}_2$ is $\min_{\mathbf{x}} \max_{\mathbf{w} \in \Delta_L} \mathbf{w}^\top \mathbf{c}(\mathbf{x})$. We approach the saddle point by alternating an inner step on $\mathbf{x}$ for fixed $\mathbf{w}$, which is the step handed to the quantum sampler, with an exponentiated-gradient step on $\mathbf{w}$ [25]:

$$\mathbf{x}^{(t)} \in \arg\min_{\mathbf{x} \text{ s.t. D1–D4}} \sum_\ell w^{(t)}_\ell\, c^{(\ell)}(\mathbf{x}), \tag{13}$$

$$\eta^{(t)} = \max_\ell c^{(\ell)}(\mathbf{x}^{(t)}), \tag{14}$$

$$w^{(t+1)}_\ell \propto w^{(t)}_\ell \exp\!\big(\mu\, c^{(\ell)}(\mathbf{x}^{(t)})/\eta^{(t)}\big), \tag{15}$$

with step $\mu > 0$ and $\mathbf{w}^{(1)}$ uniform, for T outer iterations; the weights move toward the scenarios that hurt the current schedule most. Equation (14) is the exact worst-case AoII of $\mathbf{x}^{(t)}$

over $\mathscr{L}$, computed in $O(L|\mathscr{U}|)$; SENTINEL returns the iterate with the smallest $\eta^{(t)}$ together with that value as its certificate. Because the inner problem (13) has a linear objective, it needs neither the epigraph variable nor the scenario slacks of a direct minimax encoding.

B. Candidate Pruning

The pre-elimination of Section IV-A removes inadmissible combinations. We further retain, for each UE, only the $\varkappa$ admissible combinations with the smallest worst-case cost $\max_\ell \psi^{(\ell)}_{imqh}$, plus the best combination at every other visible node so that the capacity coupling is preserved, and denote the result $\mathscr{C}_i \subseteq \mathscr{A}_i$. The number of decision qubits per UE is then at most $\varkappa + |\mathscr{M}|$ instead of $\sum_m Q_m H_i$. Pruning could in principle discard an optimal combination that is individually poor but relieves a congested node; with $\varkappa = 4$ this did not happen on any of the instances of Section VI, where the exact optimum over $\mathscr{C}$ equalled the exact optimum over $\mathscr{A}$ from 10 to 100 UEs.

C. Compact Domain QUBO

For domain k with weights $\mathbf{w}$ and ADMM state $(\mathbf{z}_k, \boldsymbol{\upsilon}_k)$, let $\bar\psi_v = \frac{1}{|\mathscr{U}|}\sum_\ell w_\ell \psi_v^{(\ell)}$ for $v = (i,m,q,h)$, $i \in \mathscr{U}_k$, $(m,q,h) \in \mathscr{C}_i$, and let $b_m(\mathbf{x}_k) = \sum_{i\in\mathscr{U}_k}\sum_{(m,q,h)\in\mathscr{C}_i} \bar\beta^q_m x_{imqh}$ be the bandwidth that domain k uses at node m. Satellites visible only to domain k form $\mathscr{S}^{\mathrm{o}}_k$; those shared with other domains form $\mathscr{S}^{\mathrm{sh}}_k$. The binary vector $\mathbf{y}_k = (\mathbf{x}_k, \boldsymbol{\nu}_k)$ holds the decision variables and, for each node $m \in \mathscr{S}^{\mathrm{o}}_k \cup \mathscr{B}_k$, a capacity slack $\nu_m = \Delta_\beta \sum_b 2^b y_{m,b}$ of $W_m = \lceil \log_2(B_m/\Delta_\beta + 1)\rceil$ bits, where the bandwidth levels and capacities are integer multiples of Δ_β. The domain QUBO is

$$F_k(\mathbf{y}_k) = \sum_v \bar\psi_v x_v + \lambda \sum_{m\in\mathscr{S}^{\mathrm{o}}_k\cup\mathscr{B}_k} \big(b_m(\mathbf{x}_k) + \nu_m - B_m\big)^2 + \frac{\lambda_{\mathrm{c}}}{2}\sum_{j\in\mathscr{S}^{\mathrm{sh}}_k}\big(b_j(\mathbf{x}_k) - z_{jk} + \upsilon_{jk}\big)^2. \quad (16)$$

The first term is the weighted AoII cost, the second enforces the capacity of the nodes that belong to the domain alone, and the third is the augmented-Lagrangian coordination term for the shared satellites, with penalty parameter λ_{c}. Constraint D1 does not appear: the mixer of Section V-D keeps every UE group one-hot, so on the subspace the circuit can reach, $x_v x_{v'} = 0$ for two candidates of the same UE and the corresponding entries of the QUBO matrix are dropped. The matrix $\mathbf{Q}^{(k)}$ collects the linear coefficients of F_k on its diagonal, since $y^2 = y$ for binary y, and the pairwise coefficients split between the two symmetric off-diagonal entries.

Proposition 1. *Let $\bar a_k = \sum_{i\in\mathscr{U}_k}\max_{v\in\mathscr{C}_i}\bar\psi_v$ and let $\Gamma^{\max}_k$ and $\Gamma^{\min}_k$ be the largest and smallest values of the coordination term over one-hot assignments of domain k. If*

$$\lambda > \frac{\bar a_k + \Gamma^{\max}_k - \Gamma^{\min}_k}{\Delta^2_\beta} \quad (17)$$

and the domain problem is feasible, then every one-hot minimizer of F_k satisfies the capacity constraints of $\mathscr{S}^{\mathrm{o}}_k \cup \mathscr{B}_k$ and minimizes the weighted cost plus coordination term over the feasible one-hot assignments.

Proof. Over one-hot assignments the cost term lies in $[0, \bar a_k]$. For a feasible assignment whose slack bits represent the unused capacity exactly, the penalty vanishes and $F_k \le \bar a_k + \Gamma^{\max}_k$. For one that exceeds the capacity of some node, the residual is a positive integer multiple of Δ_β for every choice of the non-negative slack, so $F_k \ge \lambda\Delta^2_\beta + \Gamma^{\min}_k > \bar a_k + \Gamma^{\max}_k$. No infeasible assignment is a minimizer, and on the feasible set F_k equals the cost plus coordination term once the slack bits cancel the penalty. □

The bound is taken over one-hot assignments because that is the subspace the circuit explores. Taking it over the whole hypercube, as a penalty analysis of a transverse-field QAOA would require, inflates λ by more than an order of magnitude and flattens the energy landscape to the point where neither simulated annealing nor QAOA finds a feasible point; this was the behaviour of the epigraph encoding and is quantified in Section VI-D. The number of qubits of domain k is

$$\hat N_k = \sum_{i\in\mathscr{U}_k}|\mathscr{C}_i| + \sum_{m\in\mathscr{S}^{\mathrm{o}}_k\cup\mathscr{B}_k} W_m, \quad (18)$$

independent of L. With the parameters of Section VI and two domains, the time-domain instances have 22, 34, 46 and 58 qubits per domain at 5, 10, 15 and 20 UEs, against 120 to 260 for the epigraph encoding; the validation instances of Section VI-D, which draw other topologies and candidate sets, have 44 and 78 qubits at 10 and 25 UEs against 213 and 353. The ratio is four to six at 10 to 100 UEs and larger below 10 UEs, where the epigraph's fixed registers dominate.

Greedy assignment. The classical reference used throughout, as warm start, as repair, and as the baseline of Section VI-D, processes the UEs of a domain in decreasing order of the spread of their candidate costs and gives each its cheapest candidate v by $\bar\psi_v$ whose node still has capacity, falling back to the cheapest candidate when none has.

D. Ising Mapping and XY-Mixer QAOA

The QUBO is mapped to an Ising Hamiltonian through the substitution $y_a = (1 - \sigma^z_a)/2$, where a indexes the $\hat N_k$ qubits of the domain and σ^z_a is the Pauli-Z operator on qubit a. Discarding additive constants,

$$H^{(k)}_C = \sum_a h_a\,\sigma^z_a + \sum_{a<a'} J_{aa'}\,\sigma^z_a\sigma^z_{a'}, \qquad h_a = -\frac{1}{2}\sum_{a'} Q^{(k)}_{aa'}, \qquad J_{aa'} = \frac{1}{2}Q^{(k)}_{aa'}, \quad (19)$$

where the sums over a' run over the symmetric matrix $\mathbf{Q}^{(k)}$. Couplings $J_{aa'}$ arise only between candidates that use the same node, and between those candidates and the node's slack bits, so the cost layer is sparse: the candidates of one UE are not coupled to each other, and candidates at different nodes are not coupled at all.

For the mixer we do not use the transverse field $\sum_a \sigma^x_a$, which would leave the one-hot subspace and force D1 back into the cost as a penalty. Instead, on each UE group we apply

a ring of XY interactions and keep the transverse field only on the slack bits [8]:

$$H_M^{(k)} = \sum_{i\in\mathcal{U}_k} \sum_{(a,a')\in \mathrm{ring}(\mathscr{C}_i)} \tfrac{1}{2}\left(\sigma_a^x\sigma_{a'}^x + \sigma_a^y\sigma_{a'}^y\right) + \sum_{a\in\mathrm{slack}} \sigma_a^x, \tag{20}$$

where ring$(\cdot)$ denotes adjacent pairs in a cyclic ordering of the group. Each XY term conserves the Hamming weight of its group, so every Trotter factor of $e^{-\mathrm{i}\theta H_M^{(k)}}$ does as well, and a state initialized in the one-hot subspace stays in it exactly, so a measured sample can violate D1 only through hardware noise. Each group is initialized either in its W state, the uniform superposition of its one-hot basis states, prepared with a linear-depth ladder of controlled rotations, or in the basis state of the greedy assignment of Section V-C, which we call the warm start. The depth-p state is

$$|\boldsymbol{\gamma},\boldsymbol{\theta}\rangle = \prod_{l=1}^{p} e^{-\mathrm{i}\theta_l H_M^{(k)}} e^{-\mathrm{i}\gamma_l H_C^{(k)}} |\psi_0\rangle, \tag{21}$$

and the angles $(\boldsymbol{\gamma},\boldsymbol{\theta})$ are optimized by a derivative-free classical optimizer on the sampled energy $\langle H_C^{(k)}\rangle$. The sampled bitstrings are decoded classically; samples that are not one-hot or exceed a capacity are discarded, and the feasible sample with the smallest value of (16) is returned as $\hat{\mathbf{x}}_k$. When $\hat{N}_k$ exceeds the capacity of the available device or simulator, the inner step uses a one-hot-preserving simulated annealer on the same QUBO.

E. Distributed Coordination

The domains are coupled through the satellites they share. For a shared satellite s_j, let $\mathscr{K}_j \subseteq \mathscr{K}$ be the K_j domains that see it. The coordinator assigns each $k \in \mathscr{K}_j$ a bandwidth target z_{jk} and keeps a scaled dual variable υ_{jk}; the local QUBOs carry the coordination term of (16). After the domains return $\hat{\mathbf{x}}_k^{(r+1)}$ in round r, the coordinator sets $\tilde{a}_{jk} = b_j(\hat{\mathbf{x}}_k^{(r+1)}) + \upsilon_{jk}^{(r)}$ and applies the sharing-form ADMM updates [9]

$$z_{jk}^{(r+1)} = \tilde{a}_{jk} - \frac{1}{K_j}\max\Big\{0, \sum_{k'\in\mathscr{K}_j} \tilde{a}_{jk'} - B_j^s\Big\}, \tag{22}$$

$$\upsilon_{jk}^{(r+1)} = \upsilon_{jk}^{(r)} + b_j\big(\hat{\mathbf{x}}_k^{(r+1)}\big) - z_{jk}^{(r+1)}, \tag{23}$$

which lower every target at an overloaded satellite by an equal share of the excess and accumulate the remaining violation in the duals. The round ends when the primal and dual residuals fall below a tolerance or after $R_{\max}$ rounds; λ_c is adapted by residual balancing.

Two details decide whether this loop produces good schedules. The targets are initialized from an uncoupled round ($\lambda_c = 0$) projected onto capacity, not from the equal share B_j/K_j, which pulls every domain toward using exactly its share of every shared satellite whether it needs it or not and left schedules 60 to 560% above the optimum at 50 to 100 UEs. And λ_c is scaled to the mean per-UE spread of the weighted costs $\bar{\psi}_v$, so that one bandwidth unit of deviation costs about one typical per-UE cost difference; a generic penalty two orders of magnitude larger makes the coordination term the only thing the inner solver sees. If no capacity-feasible network schedule exists when the loop stops, a greedy repair moves UEs from overloaded satellites to the candidate with the smallest increase in worst-case cost at a node with spare capacity (Algorithm 1).

Algorithm 1: SENTINEL: Distributed Robust Orchestration for VANGUARD

Input : Network, sources, scenario set $\mathscr{L}$, $\varkappa$, T, μ, p, $R_{\max}$, ϵ_{tol}.
Output: $\mathbf{x}^\star$ decoded into ρ_i, α_{ij}, ω_{in}, β_{ij}^{us}, β_{in}^{ub}; certificate $\eta^\star$; worst scenario $\ell^\star$.

Partition the network into K domains; tabulate $\Psi_i(\cdot)$ by (7), the costs $\psi_{imqh}^{(\ell)}$ by (11), and the candidates $\mathscr{C}_i$.
$\mathbf{w}^{(1)} \leftarrow$ uniform; $\eta^\star \leftarrow +\infty$.
for $t = 1, \ldots, T$ **do**
  Warm start: solve each domain with $\lambda_c = 0$; set $z_{jk}^{(0)}$ to the projected loads, $\upsilon_{jk}^{(0)} \leftarrow 0$; scale λ_c to the cost spread.
  for $r = 0, \ldots, R_{\max} - 1$ **do**
    for *each domain* k in parallel **do**
      Build F_k by (16) with $\mathbf{w}^{(t)}$, $\mathbf{z}_k^{(r)}$, $\boldsymbol{\upsilon}_k^{(r)}$; map to $H_C^{(k)}$ by (19).
      if $\hat{N}_k$ *fits the sampler* **then** run XY-mixer QAOA (21), decode, keep the best feasible sample as $\hat{\mathbf{x}}_k^{(r+1)}$;
      else obtain $\hat{\mathbf{x}}_k^{(r+1)}$ by one-hot-preserving annealing of F_k;
    Update $z_{jk}^{(r+1)}$, $\upsilon_{jk}^{(r+1)}$ by (22)–(23); adapt λ_c.
    if $\hat{\mathbf{x}}^{(r+1)}$ *satisfies D1–D3 and* $\max_\ell c^{(\ell)}(\hat{\mathbf{x}}^{(r+1)}) < \eta^\star$ **then**
      $\mathbf{x}^\star \leftarrow \hat{\mathbf{x}}^{(r+1)}$; $\eta^\star \leftarrow \max_\ell c^{(\ell)}(\mathbf{x}^\star)$.
    if *residuals* $\le \epsilon_{\mathrm{tol}}$ **then break**;
  if *no feasible schedule in this iteration* **then** apply greedy repair and evaluate (14);
  $\mathbf{x}^{(t)} \leftarrow$ best schedule of this iteration; update $\mathbf{w}^{(t+1)}$ by (15).
Decode $\mathbf{x}^\star$; $\ell^\star \leftarrow \arg\max_\ell c^{(\ell)}(\mathbf{x}^\star)$.

Convergence and certification. For convex local problems the sharing form of ADMM converges [9]; here the local problems are binary, so the loop is a heuristic and neither it nor the outer weight iteration is guaranteed to reach the saddle point of (12). SENTINEL does not rely on either for correctness. The incumbent is replaced only by network schedules that satisfy D1 to D3, and its worst-case AoII (14) is computed exactly, so the number of iterations affects the quality of the returned schedule but never the validity of its certificate.

F. Resource Scaling and Complexity

Qubits and gates. By (18) the qubit count of a domain grows with $\varkappa|\mathscr{U}_k|$ and the slack registers and does not depend on L. Because the cost Hamiltonian couples only candidates that share a node, the cost layer of domain k needs at most $\sum_m \binom{n_m+W_m}{2}$ two-qubit ZZ rotations, with n_m the number of

candidates at node m in the domain and $W_m = 0$ for shared satellites; the XY mixer adds $|\mathscr{C}_i|$ two-qubit rotations per UE group and layer and the W-state preparation $|\mathscr{C}_i| - 1$. Routed on a heavy-hex processor at depth 1, domains of 18 to 24 qubits took 58 to 305 two-qubit gates and a 38-qubit domain 606 (Section VI-E); the epigraph encoding, whose scenario penalties couple every pair of local variables, needs $\hat{N}_k(\hat{N}_k - 1)/2$ per layer with $\hat{N}_k$ four to six times larger.

Time. Tabulating Ψ_i costs $O(Y^{\max}|\mathcal{X}_i|^3 + |\mathcal{X}_i|(Y^{\max})^2)$ per UE and the coefficients $O(L\sum_i|\mathscr{A}_i|)$ lookups, once per set of sources and scenarios; assembling F_k costs $O(\hat{N}_k^2)$, the ADMM updates $O(\sum_j K_j)$ per round, and the certificate $O(L|\mathcal{U}|)$. With I_{QAOA} optimizer iterations of N_{shot} shots and the domains in parallel, a solve costs

$$C_{\mathrm{SENTINEL}} = O\Big(T\,R_{\max}\big(I_{\mathrm{QAOA}}\,p\,N_{\mathrm{shot}}\max_k G_k + L|\mathcal{U}|\big)\Big). \quad (24)$$

where G_k is the gate count above. In the evaluation a 20-UE solve with $T = 6$ and $R_{\max} = 20$ took 1 to 7 s on the classical annealer, and the orchestration loop used 34 to 223 s of solver time per ten-minute run, which enters the AoII through the overhead ϵ_i of Section III-G. Exhaustive search over $\mathscr{P}_2$ costs $O(L|\mathcal{U}|\prod_i|\mathscr{A}_i|)$, branch-and-cut on $\mathscr{P}_3$ has the same worst case although it is fast on our instances, and a centralized QAOA would need $\sum_k \hat{N}_k$ qubits on one device; the advantage of the decomposition is the qubit count per device, not asymptotic time.

VI. Performance Evaluation

We evaluate SENTINEL with the simulator released with this paper[1] under the parameters of Table 2, which are the values used for every figure. Each configuration runs for 60,000 slots of 10 ms, ten minutes of network time, on 30 independent seeds; the policies of one seed share its topology, source trajectories, fading, and disruption realization, so all comparisons are paired. We report medians with interquartile ranges, because a few seeds contain long non-line-of-sight passes that dominate means; means with 95% confidence intervals, the raw per-seed results, and the paired test tables are in the repository.

Baselines. *Random* transmits with probability 0.1 per slot over a uniformly chosen visible node with equal bandwidth sharing. *AoII threshold* is a state-aware policy that transmits whenever its AoII is positive, at most once per 5 slots, on the node with the best nominal rate. *Whittle index* is the restless-bandit scheduler of the AoII literature [11], [12]: the index of each UE's (source state, estimate, age) arm is computed numerically per source by relative value iteration on a grid of transmission prices, and every slot the mismatched UEs are ranked by index and admitted at their best node while the node's shared bandwidth still delivers every admitted packet; it has the threshold policy's observations and rate cap. *PPO* is an actor-critic agent [27] that selects, once per orchestration interval, a serving node and an update-period level for every UE, with invisible or failed nodes masked and equal bandwidth sharing, rewarded with the negative mean AoII of the interval; it is trained for 5000 episodes per network size on 30 seeds disjoint from the evaluation seeds and evaluated deterministically on the same episodes as every other policy. It acts once per interval with equal sharing, like SENTINEL, so that the comparison isolates the decision rule from the actuation; a learned per-slot policy that observes the source would be a different baseline, and that role is filled by the threshold and Whittle policies. *Nominal SENTINEL* runs Algorithm 1 with the nominal scenario alone ($L = 1$). *SENTINEL* is the robust schedule with the five-scenario set. *SENTINEL + trigger* uses SENTINEL's association and bandwidth with transmissions triggered by the observed mismatch, rate-capped at the chosen period. Every policy receives the same acknowledgement feedback and applies the same link-failure rule of Section III-F. The exact reference for Section VI-D is the integer linear program $\mathscr{P}_3$ solved with HiGHS [35].

TABLE 2. Simulation Parameters

Parameter	Value
Network	
Number of UEs	$\{5, 10, 15, 20\}$; up to 100 in Sec. VI-D
BSs / satellite serving slots	2 / 2 (topology study: 0 to 3 each)
Area; BS height; UE height	20×20 km^2; 25 m; 1.5 m
Satellite passes	Starlink TLE snapshot, SGP4, MTSU site (35.85°N, 86.39°W); min. elevation 25°
Channel and resources	
UE transmit power	23 dBm
UE–BS link	3.5 GHz, 20 MHz, $-147.55 + 20\log_{10} f + 26\log_{10} d + 6$ dB, gain 15 dBi, NF 7 dB, SE cap 6
UE–SAT link	12 GHz, 50 MHz, free space +1 dB, gains 30 + 30 dBi, NF 9 dB, SE cap 4
NTN large-scale model	TR 38.811 suburban (LOS prob., clutter, shadow fading per pass); atm. 0.5 dB/sin V
Bandwidth levels $\bar{\beta}_m^q$ ($Q_m = 3$)	$\{2.5, 5, 10\}$ MHz, $\Delta_\beta = 2.5$ MHz
Source and AoII	
Source states; transition matrix	4; birth–death, self-transition $\mathcal{U}[0.90, 0.99]$ per UE
Distortion; AoII budget $\Delta_i^{II,\max}$	$\lvert x - \hat{x}\rvert$; 10
Update periods ($H_i = 4$); slot; packet	$\{5, 10, 20, 50\}$ slots; 10 ms; 20 kbit
Disruption	
Scenario set ($L = 5$)	nominal; deep shadowing 12 dB for 30 slots; handover of each SAT (20 slots); joint loss (30 slots)
Realized process (per 1 s interval)	TR 38.811 state per pass; shadowing event per link w.p. 0.07; handover failure per SAT w.p. 0.03
Link-failure handling (all policies)	node blocked after 60 failed slots, for 6 s or until its next handover
Solver	
Formulation	compact dual, $\varkappa = 4$, $T = 6$ outer iterations, $\mu = 2$
Local solver	XY-QAOA, depth 1, 512 shots, COBYLA 40 it. ($\leq$ 26 qubits); one-hot annealing otherwise
ADMM $\lambda_{c,0}$, tolerance, rounds	scaled to cost spread, 10^{-2}, 20
Quantum backend	Qiskit Aer [36]; FakeTorino / FakeFez; `ibm_kingston`, `ibm_fez`

A. AoII Versus Network Size

Figure 2 and Table 3 give the median average AoII from 5 to 20 UEs. In every network-level result of this section the inner solver of Algorithm 1 was the one-hot-preserving annealer, because the domains span 12 to 76 qubits and are

[1] https://github.com/tanzimulalam/Quantum_SAGIN_AoII_Updated_Simulation, tag `v2.3.4-manuscript`; the script `scripts/reproduce_all.sh` regenerates every table and figure.

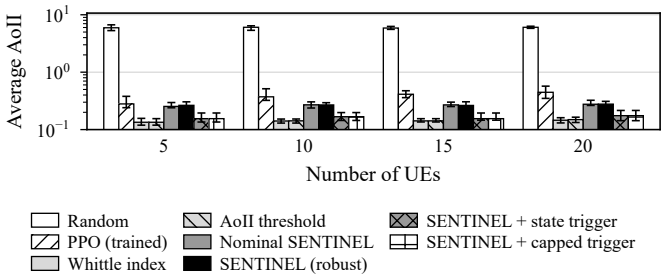


Fig. 2. Median average AoII versus the number of UEs (whiskers: IQR over 30 seeds).

TABLE 3. Median Average AoII Versus Number of UEs (30 Seeds)

UEs	Random	PPO	Whittle	AoII thr.	Nominal	SENTINEL	+ trigger
5	5.94	0.280	0.134	0.134	0.253	0.265	0.155
10	6.06	0.372	0.141	0.142	0.270	0.269	0.168
15	5.87	0.413	0.142	0.142	0.269	0.264	0.155
20	6.05	0.446	0.146	0.149	0.276	0.278	0.175

re-solved hundreds of times per run; the XY-mixer QAOA is validated against it and the exact optimum on domains of up to 24 qubits (Section VI-D) and run on hardware (Section VI-E). Three groups are visible. The state-aware per-slot policies, AoII threshold and Whittle index, are lowest at 0.13 to 0.15 and coincide (Section VI-C). The SENTINEL schedules, robust and nominal, sit at 0.25 to 0.28 and do not grow with the number of UEs, because bandwidth levels are dedicated and a re-solve redistributes load as satellites come and go. SENTINEL with the state-aware trigger lies between the two at 0.155 to 0.175: it keeps the robust association and bandwidth but transmits only when the estimate is wrong, raising the informative-update ratio from 0.23 to 0.96. Random scheduling is at 6; the trained PPO agent is close to the nominal schedule at 5 UEs (0.280 against 0.253, 11% worse), degrades to 0.446 at 20 UEs as its action space grows, routes all traffic to satellites, and, like SENTINEL, samples without observing the source. Delivery ratios are 0.92 to 0.94 for every policy except random; the per-slot policies deliver in 2.8 to 3.0 ms against 4.7 to 5.8 ms for the SENTINEL schedules, whose dedicated bandwidth levels are smaller than an equal share of an idle node. The satellite tier carries 73 to 79% of robust SENTINEL's deliveries and 97 to 98% of the threshold policy's; peak AoII is 100 to 400 for every policy, set by handovers and non-line-of-sight passes rather than by the scheduler.

B. The Certificate and the Value of Robustness

Figure 3 plots, for every run, the time-weighted certified worst case against the realized average AoII; the per-epoch statistic is reported in the text. The realized average stayed within the certificate in 98.7% of orchestration epochs for the robust schedule and 98.5% for the nominal one; the violations are passes that enter the non-line-of-sight state, whose 17 to 25 dB clutter loss exceeds the 12 dB deep-shadowing scenario. The certificate is valid against the scenario set it was computed from, and it is conservative: the certified worst case exceeds the realized average by a factor of 12 to 20 (median 15.5 over

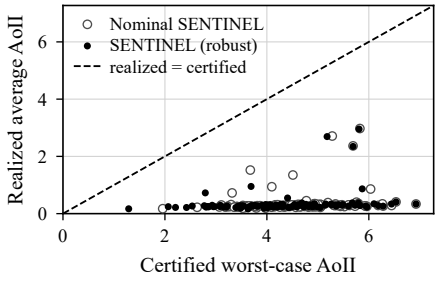


Fig. 3. Certified worst-case AoII versus realized average AoII per orchestration run, robust and nominal SENTINEL.

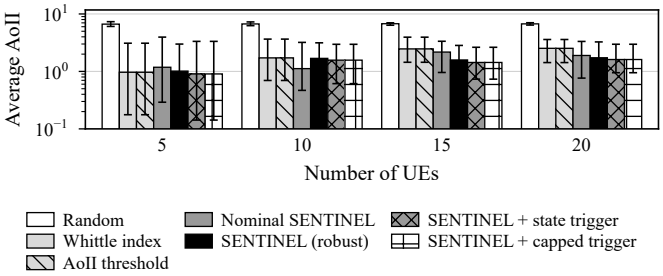


Fig. 4. Median average AoII without acknowledgement feedback or link-failure handling (60 seeds).

runs), because every scenario applies its outage to every inter-delivery interval of the epoch, which amounts to a persistent disruption, whereas realized disruptions are intermittent. Per epoch the certified-to-realized ratio has median 17 (IQR 13 to 23) over 3975 epochs, and the 1.3% of violating epochs are not marginal: a blocked pass pushes realized AoII to 1.2 to 2.1 times the certificate at the median and up to 5.4 times, so the certificate is a guarantee against the scenario set, not a prediction of typical behaviour. The certificate is proved for the periodic schedule, whose deliveries do not depend on the source state; the triggered variant transmits on observed mismatch, which the renewal argument of Section III-D does not cover, so we check it empirically: its realized average stayed within the certificate of its schedule in 97.9 to 98.7% of epochs.

What the scenario set buys is visible in the certificate, not in the realized AoII. Across the 30 paired seeds the robust schedule's certified worst case is 2 to 7% below the nominal schedule's (paired-median ratio 0.93 at 5 UEs to 0.98 at 20 UEs), lower in 67 to 100% of seeds, with Wilcoxon signed-rank $p \leq 9 \times 10^{-5}$ at every size after Holm correction. The realized AoII of the two is statistically indistinguishable at every size ($p \geq 0.08$), although the robust mean at 20 UEs is lower, 0.39 against 0.46, because the robust schedule keeps 21 to 27% of deliveries on base stations. The gap between SENTINEL with the trigger and the threshold policy, 0.02 to 0.03 in AoII, is small but significant at every size ($p \leq 0.013$), and PPO is significantly behind SENTINEL from 10 UEs on.

Without failure feedback. Whether the network can react to a dead link matters more than any scheduling choice. Figure 4 repeats the sweep with the link-failure rule disabled for every policy, on 60 seeds because the distributions become

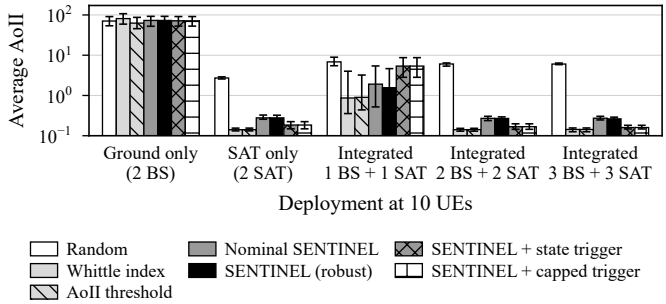


Fig. 5. Median average AoII at 10 UEs by deployment (log scale; whiskers: IQR over 30 seeds).

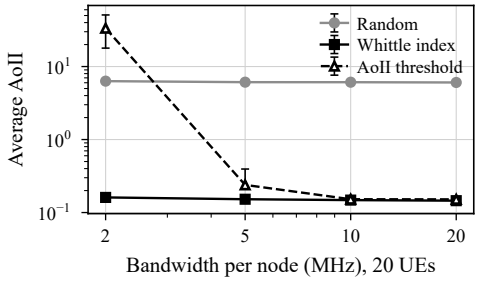


Fig. 6. Median average AoII of the Whittle-index and AoII-threshold policies at 20 UEs versus bandwidth per node.

heavy-tailed: median AoII rises by an order of magnitude for all policies, the certificate holds in 87 to 92% of epochs, and the threshold and Whittle-index policies, which coincide here as well and concentrate 98% of their traffic on the best nominal satellite, tie the SENTINEL schedules at 5 and 10 UEs and fall 25 to 30% behind SENTINEL with the trigger at 15 and 20 UEs (Hodges-Lehmann differences −0.77 and −0.71, Holm-corrected $p = 0.023$ and 0.029 against the threshold policy and 0.034 and 0.043 against the Whittle policy), while the SENTINEL schedules keep 18 to 27% of deliveries on base stations. Robust and nominal SENTINEL remain indistinguishable in realized AoII here as well ($p \geq 0.41$). The measurable product of the robust scenario set is thus its lower certificate, obtained in both regimes, and the measurable value of a scheduled base-station share appears when feedback is absent.

Scenario-set size. With the list extended by a mild 6 dB shadowing scenario and truncated to $L = 1, \ldots, 6$ at 10 UEs, realized AoII is flat in L (0.265 to 0.278), and the certificate and the base-station hedge move only once the handover of the second shared satellite enters the set (certified worst case 4.57 to 4.28, satellite share 0.79 to 0.74 between $L = 4$ and $L = 5$). The mild scenario never binds: removing it leaves every schedule's certificate unchanged on all 30 seeds at every size, which is why the paper's set has five members.

C. Topology, the Whittle Index, and Capacity

Figure 5 compares deployments at 10 UEs. Two base stations cover only the UEs within 5 to 7 km of the 20×20 km^2 area, so ground-only monitoring fails for every policy (median AoII 63 to 81). Two satellite serving slots alone bring the per-slot policies to 0.14 and the SENTINEL schedules to 0.18 to 0.28, and the integrated 2 + 2 and 3 + 3 deployments are the best for every policy, with SENTINEL routing 26 to 36% of its deliveries to base stations. The one deployment in which the SENTINEL schedules are clearly worse than the per-slot policies is a single base station with a single satellite slot (medians 1.6 to 1.9 against 0.87 to 0.90, and 5.4 for the triggered hybrid): with one slot every handover is a total outage for every UE on it, and a scheduled policy loses more during the recovery than one that retransmits the moment the link returns; this is a limitation of scheduled transmission at that topology.

TABLE 4. Mean Optimality Gap (%) Above the Exact ILP Optimum of $\mathscr{P}_3$

Setting	Qubits/domain	XY-QAOA	Compact ADMM	Epigraph ADMM	Greedy+LS
4–8 UEs, 2+2	8–24	0.0 (15/15 optimal)	0.0	38–51	0.0
10 / 25 UEs, 2+2	44 / 78	–	0.0	42.7 / 1.8	0.0
50 UEs, 4+3	142	–	11.5	38.8	6.0
75 / 100 UEs	159 / 205	–	49.9 / 60.5	–	14.4 / 11.5
Severity sweep, 20 UEs	29–66	–	0.0	0.2–4.7	0.0
$L = 1/3/5/6$, 20 UEs	60–77	–	0.0/3.2/6.9/0.0	74/233/9.0/3.8	0/0/1.1/0

Whittle index and capacity. At the bandwidths of Table 2 a node delivers 19 to 40 simultaneous equal sharers within one slot, so capacity never binds and the Whittle scheduler makes the same decisions as the threshold policy (identical AoII to three decimals at every size and topology, $p \geq 0.18$); the threshold baseline of this paper is the Whittle-index policy of this regime. Figure 6 reduces the bandwidth per node at 20 UEs: at 5 MHz the threshold policy is 57% worse than the index scheduler (0.239 against 0.152, $p = 4 \times 10^{-9}$), and at 2 MHz its simultaneous transmissions collapse every shared link (median AoII 33.5, delivery ratio 0.16) while the capacity-aware scheduler stays at 0.16; SENTINEL's dedicated bandwidth levels avoid this regime by construction. Its one weakness is the ground-only deployment (81 against 63), where its admission rule never attempts a node that cannot deliver at the nominal rate.

D. Validation Against the Exact Minimax Optimum

Table 4 reports the mean gap to the exact ILP optimum of $\mathscr{P}_3$ over capacity-feasible schedules (10 seeds; 5 for the QAOA rows), for the XY-mixer QAOA on the statevector simulator, for the full distributed pipeline of Algorithm 1 with the compact QUBO and with the epigraph encoding of our earlier version, and for greedy association refined by local search. The XY-mixer QAOA returned the exact optimum on every instance whose domains fit the simulator (8 to 24 qubits), with 100% of samples one-hot and capacity-feasible; the epigraph encoding, run with the same classical search, is 38 to 51% above the optimum on the same instances with 133 to 193 qubits, and a plain simulated annealer on its penalty landscape found no feasible assignment in 50 solves at 10 UEs. The compact distributed pipeline matches the exact optimum on all 10- and 25-UE instances and on the severity instances at 20 UEs, is within 3 to 7% on the 20-UE scenario-count instances (optimal on 60 to 90% of them), is within 12% at 50 UEs, and is limited at 75 to 100 UEs by the coordination of five shared satellites among six to eight domains, which

TABLE 5. Device Sampling Results on IBM Heron Processors (greedy start: `ibm_kingston`, seeds 6 and 7 on `ibm_fez`; W-state start on the same processor; U: number of UEs, s: seed, d: domain; uniform baseline per run)

Instance	Qubits	2q gates		Spread	Feasible shots		P(opt. | feas.)		Uniform P(opt.)	
		greedy	W		greedy	W	greedy	W	greedy	W
U6 s1 d0	18	139	179	43%	63%	62%	0.95	0.17	0.125	0.125
U6 s1 d1	20	301	314	307%	43%	36%	0.93	0.15	0.125	0.125
U6 s2 d0	18	139	179	35%	61%	62%	0.95	0.09	0.031	0.125
U6 s2 d1	12	58	78	0%	74%	74%	0.94	0.12	0.016	0.125
U6 s3 d0	24	305	333	115%	35%	34%	0.91	0.03	0.062	0.062
U6 s3 d1	14	120	154	263%	70%	74%	0.96	0.20	0.250	0.250
U6 s5 d0	12	58	82	0%	76%	73%	0.98	0.13	0.125	0.125
U6 s5 d1	12	58	82	0%	76%	74%	0.98	0.12	0.125	0.125
U6 s6 d0	8	38	52	1%	86%	65%	0.99	0.33	0.250	0.250
U6 s6 d1	16	79	108	1%	62%	44%	0.95	0.13	0.062	0.062
U6 s7 d0	12	58	80	2%	72%	59%	0.97	0.12	0.125	0.125
U6 s7 d1	18	142	185	63%	59%	49%	0.93	0.07	0.125	0.125
U10 s1 d0	38	606	719	82%	3%	3%	0.24	0.00	0.006	0.006

neither more ADMM rounds nor smaller dual steps resolve and where greedy association with local search stays within 11 to 14%. Pruning with $\varkappa = 4$ never changed the optimum. The exact reference was solved to proven optimality in under a second up to 75 UEs and hit its time limit on one in ten instances at 100 UEs.

E. Quantum Resources and Device Results

Table 5 reports the sampling of the hardware-sized domains on the `ibm_kingston` and `ibm_fez` processors (Heron r2, 156 qubits; $\varkappa = 4$, depth 1, 4096 shots per domain, transpiled at optimization level 3 with eight routing trials; 27 October to 1 November 2026). Every scenario weight, ADMM target, and QAOA angle was optimized on the simulator and each device job executed only the final sampling circuit of one or two domains; the 38-qubit domain exceeds the statevector simulator and received the angles of the largest simulated domain of its run, relying on parameter concentration [32]. Per run the table lists the feasible-shot fraction, the probability of the optimum among the feasible shots, and the same probability under uniform sampling of the run's feasible one-hot set, which we enumerate; the product of the first two is the probability that a single shot is an optimal assignment. The energy spread of the feasible set is listed because several hardware-sized domains are nearly flat: where it is below 2%, any feasible assignment is within 2% of the optimum and the sampler only has to achieve feasibility; on the other domains it is 35 to 307%, and uniform sampling reaches an approximation ratio of only 0.62 to 0.91.

Three facts follow from the table. First, from the greedy warm start, which is itself optimal on every hardware-sized domain, the device returned an optimal assignment in 32 to 85% of all shots on the 6-UE domains, 91 to 99% of the feasible shots, against 1.6 to 25% for uniform feasible sampling. This measures the retention of an already-optimal warm start under noise, not a search, and we state it that way. Second, at 38 qubits the per-shot probability of the optimum, 0.6%, equals the uniform baseline: noise destroys 97% of the shots, and only among the surviving feasible shots does the warm start still show (24% optimal against 0.6%). The best feasible sample equalled the exact optimum of its domain in every run and every decoded schedule attained the ILP optimum of its instance, but at 38 qubits this is what 4096 uniform feasible samples would also achieve. Third, from the W state, with no warm start, run on the same processor as the greedy start, the device is a uniform sampler of the feasible set: the probability of the optimum among feasible shots is 0.03 to 0.33 against the uniform 0.06 to 0.25 of the same run (each run re-solves the scenario weights, and on the two seed-2 domains this changes the optimal set, so the table lists the baseline per run), between 0.52 and 2.04 times the uniform value on every 6-UE domain, and the approximation ratio of feasible shots exceeds the uniform value by at most 0.07; at 38 qubits no feasible shot was optimal. The 1.3-fold concentration of one noiseless layer is lost to hardware noise. Every decoded schedule still attained the ILP optimum of its instance (at 38 qubits the best domain sample was suboptimal but tied at the network worst case), because the feasible sets are small and the decoder keeps the best of thousands of shots.

The feasible-shot fraction measures the noise: 59 to 86% at 8 to 18 qubits with up to 142 two-qubit gates, 35 to 43% at 20 to 24 qubits with about 300 gates, and 2.6% at 38 qubits with 606 gates, bracketed by the FakeTorino and FakeFez noise models (52/33% and 70/48% on the 18- and 20-qubit domains of the first instance); the W-state circuits, 13 to 113 gates deeper for the state preparation, give the same fractions on the same processor (34 to 74% and 3.5% on `ibm_kingston`). A few hundred more two-qubit gates would leave fewer than one feasible shot in a thousand, which places the present limit of the quantum step at about ten UEs per domain at depth 1. Error suppression does not move it: re-sampling the 24- and 38-qubit domains in one calibration session with dynamical decoupling (XY4) [33], and with dynamical decoupling plus Pauli twirling [34], lowered the feasible fraction from the in-session control values of 25.0% and 2.4% to 21.1% and 0.7%, and to 13.8% and 0.2%, because these circuits have almost no idle time in which decoupling could help and twirling inserts further gates. Because the unmitigated yield of one circuit varied by ten points between calibration sessions (34.8% against 25.0%), all device comparisons are made within one session.

We therefore claim no quantum advantage: in simulation the QAOA layers only concentrate a constrained sampler (0.42 at depth 2 from the W state), and on every hardware-sized domain greedy association is already optimal. The instances on which it is not, 50 UEs and above in Table 4, need 140 to 200 qubits per domain, beyond simulation and beyond the gate budget that the feasible-shot fractions above allow on current devices; whether a sampler at that size concentrates on the optimal set is the question this work leaves open.

VII. Conclusion

This paper asked how a satellite-ground edge network should schedule status updates when what matters is whether the edge estimate is correct, and when the satellite links that carry most of the traffic fail for a fraction of a second at each handover and for a whole pass under blockage. We formulated the robust problem VANGUARD, reduced the history-dependent AoII to a tabulated cost per inter-delivery in-

terval, and solved the resulting minimax assignment with SENTINEL, which dualizes the minimax into classically updated scenario weights, encodes each domain's linear subproblem as a compact QUBO whose one-hot structure the XY mixer preserves, coordinates the domains by ADMM, and certifies every schedule by its exact worst case.

Under Starlink passes and the TR 38.811 channel the certificate held in 98.7% of epochs; the robust schedule's certificate was 2 to 7% below the nominal one while the two were indistinguishable in realized AoII with and without failure feedback; and SENTINEL with a state-aware trigger came within 10 to 20% of the best per-slot heuristic, which coincides with the Whittle-index scheduler at these bandwidths, and beat it by 25 to 30% at 15 to 20 UEs without feedback. The compact formulation needs four to six times fewer qubits than a direct epigraph encoding, matches the exact optimum on the 10- and 25-UE instances, and is limited at 75 to 100 UEs by the coordination of shared satellites; thirteen device domains of 8 to 38 qubits decoded to the exact optimum, with the fraction of usable shots falling from 86% to 2.6% and not improved by error suppression, and at these sizes the quantum sampler is correct but not a better optimizer than greedy association. Three directions follow: a coordinator that negotiates per-UE assignments rather than aggregate bandwidth, to remove the bottleneck above 50 UEs; a scenario set learned from the pass statistics of the site; and whether a constrained quantum sampler concentrates on the optimal set at the 150-qubit scale where greedy association starts to fail.

REFERENCES

[1] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134–142, May 2020.

[2] M. Giordani and M. Zorzi, "Non-terrestrial networks in the 6G era: Challenges and opportunities," *IEEE Netw.*, vol. 35, no. 2, pp. 244–251, Mar. 2021.

[3] 3GPP, "Study on New Radio (NR) to support non-terrestrial networks," 3rd Generation Partnership Project, Tech. Rep. TR 38.811, Oct. 2020.

[4] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *Proc. IEEE INFOCOM*, Orlando, FL, USA, Mar. 2012, pp. 2731–2735.

[5] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 5, pp. 1183–1210, May 2021.

[6] A. Maatouk, S. Kriouile, M. Assaad, and A. Ephremides, "The age of incorrect information: A new performance metric for status updates," *IEEE/ACM Trans. Netw.*, vol. 28, no. 5, pp. 2215–2228, Oct. 2020.

[7] A. Maatouk, M. Assaad, and A. Ephremides, "The age of incorrect information: An enabler of semantics-empowered communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2621–2635, Apr. 2023.

[8] S. Hadfield, Z. Wang, B. O'Gorman, E. G. Rieffel, D. Venturelli, and R. Biswas, "From the quantum approximate optimization algorithm to a quantum alternating operator ansatz," *Algorithms*, vol. 12, no. 2, p. 34, 2019.

[9] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.

[10] D. A. Vallado, P. Crawford, R. Hujsak, and T. S. Kelso, "Revisiting Spacetrack Report #3," in *Proc. AIAA/AAS Astrodynamics Specialist Conf.*, Keystone, CO, 2006, aIAA 2006-6753.

[11] P. Whittle, "Restless bandits: Activity allocation in a changing world," *J. Appl. Probab.*, vol. 25, no. A, pp. 287–298, 1988.

[12] I. Kadota, A. Sinha, E. Uysal-Biyikoglu, R. Singh, and E. Modiano, "Scheduling policies for minimizing age of information in broadcast wireless networks," *IEEE/ACM Trans. Netw.*, vol. 26, no. 6, pp. 2637–2650, Dec. 2018.

[13] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 2017, 4th Quart.

[14] J. Liu, Y. Shi, Z. M. Fadlullah, and N. Kato, "Space-air-ground integrated network: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2714–2741, 2018, 4th Quart.

[15] N. Kato, Z. M. Fadlullah, F. Tang, B. Mao, S. Tani, A. Okamura, and J. Liu, "Optimizing space-air-ground integrated networks by artificial intelligence," *IEEE Wireless Commun.*, vol. 26, no. 4, pp. 140–147, Aug. 2019.

[16] Z. Zhang, W. Zhang, and F.-H. Tseng, "Satellite mobile edge computing: Improving QoS of high-speed satellite-terrestrial networks using edge computing techniques," *IEEE Netw.*, vol. 33, no. 1, pp. 70–76, Jan. 2019.

[17] N. Cheng, F. Lyu, W. Quan, C. Zhou, H. He, W. Shi, and X. Shen, "Space/aerial-assisted computing offloading for IoT applications: A learning-based approach," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 5, pp. 1117–1129, May 2019.

[18] C. Zhou, W. Wu, H. He, P. Yang, F. Lyu, N. Cheng, and X. Shen, "Deep reinforcement learning for delay-oriented IoT task scheduling in SAGIN," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 911–925, Feb. 2021.

[19] A. Kosta, N. Pappas, and V. Angelakis, "Age of information: A new concept, metric, and tool," *Found. Trends Netw.*, vol. 12, no. 3, pp. 162–259, 2017.

[20] Y. Sun, E. Uysal-Biyikoglu, R. D. Yates, C. E. Koksal, and N. B. Shroff, "Update or wait: How to keep your data fresh," *IEEE Trans. Inf. Theory*, vol. 63, no. 11, pp. 7492–7508, Nov. 2017.

[21] S. Kriouile and M. Assaad, "Minimizing the age of incorrect information for real-time tracking of Markov remote sources," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Jul. 2021, pp. 2978–2983.

[22] Y. Chen and A. Ephremides, "Minimizing age of incorrect information for unreliable channel with power constraint," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2021, pp. 1–6.

[23] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski, *Robust Optimization*. Princeton, NJ, USA: Princeton Univ. Press, 2009.

[24] D. Bertsimas and M. Sim, "The price of robustness," *Oper. Res.*, vol. 52, no. 1, pp. 35–53, Feb. 2004.

[25] S. Arora, E. Hazan, and S. Kale, "The multiplicative weights update method: A meta-algorithm and applications," *Theory Comput.*, vol. 8, no. 6, pp. 121–164, 2012.

[26] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Feb. 2015.

[27] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[28] A. Lucas, "Ising formulations of many NP problems," *Front. Phys.*, vol. 2, p. 5, Feb. 2014.

[29] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," arXiv:1411.4028, 2014.

[30] T. Albash and D. A. Lidar, "Adiabatic quantum computation," *Rev. Mod. Phys.*, vol. 90, no. 1, p. 015002, Jan. 2018.

[31] M. Kim, D. Venturelli, and K. Jamieson, "Leveraging quantum annealing for large MIMO processing in centralized radio access networks," in *Proc. ACM SIGCOMM*, Beijing, China, Aug. 2019, pp. 241–255.

[32] F. G. S. L. Brandão, M. Broughton, E. Farhi, S. Gutmann, and H. Neven, "For fixed control parameters the quantum approximate optimization algorithm's objective function value concentrates for typical instances," 2018, arXiv:1812.04170.

[33] L. Viola, E. Knill, and S. Lloyd, "Dynamical decoupling of open quantum systems," *Phys. Rev. Lett.*, vol. 82, no. 12, pp. 2417–2421, 1999.

[34] J. J. Wallman and J. Emerson, "Noise tailoring for scalable quantum computation via randomized compiling," *Phys. Rev. A*, vol. 94, no. 5, p. 052325, 2016.

[35] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Math. Program. Comput.*, vol. 10, no. 1, pp. 119–142, 2018.

[36] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, "Quantum computing with Qiskit," 2024, arXiv:2405.08810.